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CANETWORK

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(Q/Ans) Real-world application of Parallel Computing and Networked Systems are -

1) Parallel Computing: Application in Weather forecasting

Use case:

Weather prediction systems, such as those run by the NWS or the IMD, use parallel computing to simulate atmospheric conditions.

How it works

1) Large-scale simulations divide the Earth's atmosphere into

2) Each grid point processes data like temperature, humidity, pressure, etc, simultaneously using high-performance computing clusters.

3) Parallel algorithms calculate data in each region concurrently to generate forecasts rapidly.

Q2) Network Systems: Application in online Multiplayer Gaming

Use Case

Popular games like Fortnite, Call of Duty and PUBG rely heavily on networked systems to connect millions of players globally.

How it Used

- 1) Game servers manage real-time interactions between users.
- 2) Client-server architecture ensures that data is exchanged instantly via the internet.
- 3) Backend services support load balancing, matchmaking, and player data sync.